

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. Local analytic response and global bounded-source Fenchel duality are kept on distinct domains. We compute the full second variation of the contact perspective entropy, including the nonlinear second derivative of the Boltzmann reference measure. A weighted linearized balance estimate proves closed range and identifies the exact gauge annihilator. Lax–Milgram, rather than formal infinite-dimensional Hessian inversion, constructs the covariance operator. Integrated deterministic-interval cumulant densities give nuclear-space Gaussian process tightness without point-conditioned stopping states.

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1. ROUND-SEVEN RECONSTRUCTION: EXACT TANGENT ACTION AND CLOSED BALANCE RANGE

This section replaces the round-six controlling module. The nonlinear reference collision measure A_f is differentiated exactly. Covariance is constructed by Lax–Milgram on the linearized balanced tangent space rather than by an unqualified infinite-dimensional Legendre inversion.

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1.1. Local analytic and global convex source domains. Let $\mathcal{S}_{\text{an}}$ be the union of bounded weighted source balls on which the B2 pressure has a zero-free complex neighborhood. It is used only for analytic response. Let

$$\mathcal{S}_{\text{cvx}} = \bigcup_{M, R < \infty} \{(p, \psi) : \text{supp}(p, \psi) \subset K_R, \|p\|_\infty + \|\psi\|_\infty \leq M\}.$$

It is used for Fenchel duality. For a general signed source there is no asserted monotone truncation limit; the global convex pressure is the directed supremum over the finite test family appearing in the dual pairing.

Theorem 1.1 (Two-domain Fenchel theorem). *For every balanced pair (f, Γ) ,*

$$I(f, \Gamma) = \sup_{(p, \psi) \in \mathcal{S}_{\text{cvx}}} \{\langle p, f \rangle + \langle \psi, \Gamma \rangle - Q(p, \psi)\}.$$

If the pair is unbalanced or if Γ has a component singular to A_f , the supremum is $+\infty$. On $\mathcal{S}_{\text{an}}$, Q is analytic and its finite derivatives agree with the corresponding finite-volume cumulants.

Proof. Compactly supported bounded sources separate Radon measures and can be scaled without leaving $\mathcal{S}_{\text{cvx}}$. Balance defects are therefore assigned infinite cost. On the absolutely continuous contact part, pointwise conjugacy of $e^z - 1$ and $\ell(q) = q \log q - q + 1$, followed by monotone spatial and velocity cutoffs, gives the entropy. A singular contact part is detected by bounded sources supported where A_f is arbitrarily small. Analyticity on the local domain follows from the B2 pressure theorem and Cauchy differentiation. $\square$

1.2. The exact second variation. For positive measures g, a write the perspective entropy density

$$\Phi(g, a) = g \log(g/a) - g + a.$$

At $g = qa$, its Hessian in affine directions (h, b) is

$$D^2\Phi_{(qa, a)}[(h, b)]^2 = \frac{(h - qb)^2}{qa}.$$

If $a_s = A_{f_s}$ is nonlinear, then

$$\left. \frac{d^2}{ds^2} \Phi(g_s, A_{f_s}) \right|_{s=0} = \frac{(\dot{\Gamma} - qDA_f[\dot{f}])^2}{qA_f} + (1 - q)D^2A_f[\dot{f}, \dot{f}].$$

This includes the term omitted in the preceding revision.

Let $I_0(f_0)$ be the prepared initial entropy. For a regular exposed phase $(f, \Gamma = qA_f)$ define

$$\mathfrak{q}_{f, q}(\dot{f}, \dot{\Gamma}) = D^2I_0(f_0)[\dot{f}_0]^2 + \int_0^T \int \frac{|\dot{\Gamma} - qDA_f[\dot{f}]|^2}{q} dA_f dt + \int_0^T \int (1 - q)D^2A_f[\dot{f}, \dot{f}] dt.$$

The tangent is required to satisfy the exact linearized balance equation.

Lemma 1.2 (Coercivity of the on-shell tangent action). *On every regular B2 analytic phase chart there are $c, C > 0$ such that*

$$\mathfrak{q}_{f,q}(\dot{f}, \dot{\Gamma}) \geq c \left(\|\dot{f}_0\|_{L^2(f_0^{-1})}^2 + \int \frac{|\dot{\Gamma} - qDA_f[\dot{f}]|^2}{q} dA_f dt + \|\dot{f}\|_{\mathsf{X}_T}^2 \right).$$

Here X_T is the weighted transport energy norm. The constants are uniform after a fixed finite horizon is divided into analytic continuation blocks.

Proof. For hard spheres, $D^2A_f[\dot{f}, \dot{f}]$ is twice the collision measure with density $\dot{f}\dot{f}_*$. Cauchy–Schwarz in (v, v_*, ω) and the Maxwellian bounds give

$$\left| \int (1-q)D^2A_f[\dot{f}, \dot{f}] \right| \leq \eta \|\dot{f}\|_{\mathsf{X}_T}^2 + C_\eta \int \frac{|\dot{\Gamma} - qDA_f[\dot{f}]|^2}{q} dA_f.$$

The linearized balance energy estimate controls X_T by the initial term and the square contact residual. On a sufficiently short analytic continuation block the coefficient η is absorbed. Concatenating finitely many blocks propagates a positive constant over the fixed horizon. $\square$

1.3. The balance operator and its closed range. Let H_{tan} be the completion of smooth tangent pairs for the norm in Lemma 1.2. Let Y be the weighted negative Sobolev space of balance defects including endpoint traces. Define

$$\mathcal{B}(\dot{f}, \dot{\Gamma}) = \partial_t \dot{f} + v \cdot \nabla_x \dot{f} - \mathcal{C}_f \dot{\Gamma},$$

with the prepared endpoint map included. Its adjoint is the backward transport–collision operator on the source pair (p, ψ) .

Theorem 1.3 (Closed balance range and the gauge annihilator). *The operator $\mathcal{B} : \mathsf{H}_{\text{tan}} \rightarrow \mathsf{Y}$ has closed range. Its kernel is the balanced tangent Hilbert space H_{bal} , and*

$$(\mathsf{H}_{\text{bal}})^\perp = \overline{\text{Ran } \mathcal{B}^*}.$$

Under the source identification, $\text{Ran } \mathcal{B}^$ is exactly the balance gauge $(r, -\Delta r)$ together with endpoint coboundaries and the prepared constraint covectors. There are no additional zero-variance source classes.*

Proof. Solve the backward adjoint equation with terminal value zero. Multiplying by the solution and using the collision Dirichlet form gives the observability estimate

$$\|(p, \psi)\|_{\mathsf{Y}^*/\ker \mathcal{B}^*} \leq C \|\mathcal{B}^*(p, \psi)\|_{\mathsf{H}_{\text{tan}}^*}.$$

The transport trace terms are controlled by the measure-flux Green identity of the B2 trace theorem. This estimate is equivalent to closed range. The Hilbert closed-range theorem gives the annihilator identity. Expanding the adjoint balance pairing yields $(r, -\Delta r)$ in the bulk/contact variables; the only boundary terms are the declared endpoint and prepared constraints. $\square$

1.4. Covariance by Lax–Milgram. For a source class ξ modulo the gauge, define $u_\xi \in \mathbf{H}_{\text{bal}}$ by

$$\mathbf{q}_{f,q}(u_\xi, v) = \langle \xi, v \rangle \quad \text{for every } v \in \mathbf{H}_{\text{bal}}.$$

Theorem 1.4 (Tangent covariance isomorphism). *The equation above has a unique solution and*

$$\text{Cov}(\xi, \eta) = \mathbf{q}_{f,q}(u_\xi, u_\eta)$$

is a positive continuous bilinear form on the quotient source Hilbert space. Its kernel is exactly the closed gauge of Theorem 1.3. Every finite-dimensional projected pressure Hessian equals the corresponding covariance matrix.

Proof. Lemma 1.2 and Lax–Milgram give existence, uniqueness, and a bounded inverse on $\mathbf{H}_{\text{bal}}$. A source has zero covariance if and only if it annihilates every balanced tangent, which by Theorem 1.3 is precisely the gauge class. Differentiating the finite-dimensional B2 variational formula twice and using the unique linearized minimizer gives the same Lax–Milgram equation, hence the projected Hessian identity. No global Frechet inversion of an arbitrary lower semicontinuous Legendre transform is invoked. $\square$

1.5. Localized deterministic-interval cumulants. For a bounded test ξ supported in an interval I , let $Y_I^\varepsilon(\xi)$ be the exactly finite-volume centered density/contact increment. Differentiating the B2 connected expansion with time-localized sources gives joint cumulant densities $c_k^\varepsilon(t_1, \dots, t_k)$. The measure-trace renewal theorem bounds these densities in L^1 uniformly on bounded tilted charts.

Lemma 1.5 (Interval and two-interval cumulant bounds). *For $k \geq 2$,*

$$|\kappa_k(Y_I^\varepsilon)| \leq C_k \left(|I| \mu_\varepsilon^{1-k/2} + \mu_\varepsilon^{-k/2} \right).$$

For adjacent disjoint intervals I, J ,

$$\mathbb{E}|Y_I^\varepsilon|^2 |Y_J^\varepsilon|^2 \leq C|I||J| + C\mu_\varepsilon^{-1}(|I| + |J|) + C\mu_\varepsilon^{-2}.$$

These are integrated deterministic-interval estimates and do not condition on a point collision.

Proof. A connected cumulant meeting a source supported in I has one distinguished marked time integrated over I . The remaining time variables are integrated against the L^1 connected density supplied by the B2 measure-trace renewal bound. Coincident endpoint atoms have total mass $O(\mu_\varepsilon^{-k/2})$. For two intervals, expand the fourth joint moment into connected cumulants and pairings. Every connected block meeting both intervals carries the product $|I||J|$; blocks meeting one endpoint give the smaller terms. $\square$

1.6. The Gaussian process. Choose a countably Hilbert nuclear test space

$$\mathbf{S} = \bigcap_{m \geq 0} H_{x,v}^m(e^{\beta_m |v|^2}) \times \bigcap_{m \geq 0} H_{\text{contact}}^m(e^{\beta_m(|v|^2 + |v_*|^2)}),$$

with Hilbert–Schmidt embeddings obtained by increasing both derivative order and Maxwellian weight. Let $\mathbf{S}'$ be its dual. The linearized biased balance generates a continuous propagator $U(t, s)$ on this scale.

Theorem 1.6 (Exact-centred density–actual-contact Gaussian limit). *The fields centered at their exact finite-volume tilted expectations converge in $D([0, T], \mathbf{S}')$ to the unique centered Gaussian solution of the linearized balance equation. Its initial covariance is the B1 constrained Schur complement and its dynamic covariance is*

$$\int_0^T \int q_t(\Delta\phi_t + \psi_t)(\Delta\chi_t + \eta_t) dA_{f_t} dt,$$

with tests propagated backward by $U(T, t)^*$. The covariance kernel is the one in Theorem 1.4.

Proof. For finitely many localized tests, second cumulants converge by the B2 pressure and higher cumulants vanish by Lemma 1.5. This gives Gaussian finite-dimensional limits. The two-interval estimate yields a uniform fractional Sobolev bound $W^{\alpha,4}$ for every $\alpha < 1/4$ after the microscopic jump correction; compact embedding gives scalar path tightness without stopping times. Mitoma’s theorem applies to the explicitly nuclear space $\mathbf{S}$. Differentiating the limiting balance at the regular phase identifies the propagator and the contact covariance. Coercivity and the linearized energy estimate give uniqueness of the Gaussian martingale problem. $\square$

Theorem 1.7 (Round-seven B3 closure). *The global entropy dual and local analytic response use distinct source domains; the nonlinear contact action has its exact second variation; the linearized balance has closed range; covariance is the Lax–Milgram inverse on the balanced quotient; and process tightness follows from integrated interval cumulants rather than an unavailable stopping trace density.*

Proof. Combine Theorems 1.1, 1.3, 1.4, and 1.6, together with Lemmas 1.2 and 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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